

EXPERIMENT 3: Performance Analysis of IIR Filters.

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB.

PROGRAM

```
clear;

close all;

%% Specifications

Fs = 4000;      % Sampling Frequency (Hz)

Rp = 1;         % Passband Ripple (dB)

Rs = 60;        % Stopband Attenuation (dB)

% Passband and Stopband Frequencies

Wp = [500 700]/(Fs/2); % Passband (500-700 Hz)

Ws = [400 800]/(Fs/2); % Stopband (400-800 Hz)

%% Butterworth Filter

[n1,Wn1] = buttord(Wp,Ws,Rp,Rs);

[b1,a1] = butter(n1,Wn1,'bandpass');

%% Chebyshev Type-I Filter

[n2,Wn2] = cheb1ord(Wp,Ws,Rp,Rs);

[b2,a2] = cheby1(n2,Rp,Wn2,'bandpass');

%% Magnitude Response

figure;

freqz(b1,a1,1024,Fs);

title('Butterworth Bandpass Filter');

figure;
```

```

freqz(b2,a2,1024,Fs);
title('Chebyshev Type-I Bandpass Filter');

%% Group Delay
figure;
grpdelay(b1,a1,1024,Fs);
hold on;
grpdelay(b2,a2,1024,Fs);
grid on;
legend('Butterworth','Chebyshev-I');
title('Group Delay');

%% Pole-Zero Plot
figure;
subplot(1,2,1);
zplane(b1,a1);
title('Butterworth Pole-Zero Plot');
subplot(1,2,2);
zplane(b2,a2);
title('Chebyshev-I Pole-Zero Plot');

%% Impulse Response
figure;
subplot(2,1,1);
impz(b1,a1);
title('Butterworth Impulse Response');
subplot(2,1,2);
impz(b2,a2);
title('Chebyshev-I Impulse Response');

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%% Pole Magnitude

figure;

subplot(1,2,1);

stem(abs(roots(a1)), 'filled');

grid on;

xlabel('Pole Number');

ylabel('|Pole|');

title('Butterworth Pole Magnitude');

subplot(1,2,2);

stem(abs(roots(a2)), 'filled');

grid on;

xlabel('Pole Number');

ylabel('|Pole|');

title('Chebyshev-I Pole Magnitude');

%% Display Filter Orders

fprintf('\n');

fprintf('Butterworth Filter Order = %d\n', n1);

fprintf('Chebyshev Type-I Filter Order = %d\n', n2);

```

OUTPUT



